The constant $A(p)$

Quasi-stationary amplitude, closed forms, and the Koenigs obstruction

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Chapter [The constant $A(p)$]

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Contents

Chapter 3

The constant $A(p)$

3.1 Introduction

The quasi-stationary theory of Chapter [Mathematical introduction]reduces the late-time conditional behaviour of the subcritical binary Galton–Watson process to a single constant

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \qquad p \in [0, \tfrac{1}{2}), \tag{3.1}$$

whose reciprocal is the limiting conditional mean. Approximate values are easy enough to obtain by iterating the survival recursion, but iteration fails as $p \to \frac{1}{2}^-$ because the number of generations needed for near-stationarity grows without bound. No formula in elementary functions is known for $A(p)$.

This chapter develops the analytic theory of that constant: an exact infinite product; an exact series with two-sided bounds; the near-critical asymptotic; the comparison with the continuous-time constant $A_c(p)$; a record of closed-form searches; the identification of $A(p)$ with a value of the Koenigs linearising function of the logistic map; and a careful account of what hypertranscendence does and does not imply about $A(p)$, together with inverse-symbolic evidence. Standing setup is recalled briefly from Chapter [Mathematical introduction]; proofs of Galton–Watson extinction and of the conditional mean and variance are not repeated.

The results of sections 3.3 to 3.5 and 3.8 are original to this thesis in the form presented here. Hypertranscendence claims for the conjugacy follow Becker–Bergweiler [1], stated and applied with the scope restrictions of section 3.10.

3.2 Standing setup

Throughout, $(Z_n)_{n\geq 0}$ denotes the binary Galton–Watson process with offspring probability generating function

$$\phi(z) = pz^2 + (1-p), \qquad p \in [0, \tfrac{1}{2}), \tag{3.2}$$

started from $Z_0 = 1$. The mean population is $\mathbb{E}(Z_n) = (2p)^n$. Writing $S_n = \mathbb{P}(Z_n > 0)$ for the survival probability, the first-step relation $S_{n+1} = 1 - \phi(1 - S_n)$ specialises to the logistic recursion

$$S_{n+1} = 2p\, S_n - pS_n^2, \qquad S_0 = 1. \tag{3.3}$$

Extinction is certain for $p \le \frac{1}{2}$ (Chapter [Mathematical introduction]). For $p < \frac{1}{2}$ the late-time survival probability admits the asymptotic

$$S_n \sim A(p)\,(2p)^n, \qquad A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \tag{3.4}$$

and the conditional mean converges to the reciprocal of this constant:

$$\mathbb{E}(Z_n \mid Z_n > 0) = \frac{(2p)^n}{S_n} \xrightarrow{n\to\infty} \frac{1}{A(p)}. \tag{3.5}$$

Existence of the limit $A(p) \in (0, \frac{1}{2}]$ is proved in section 3.3 via the infinite product. The full derivation of the conditional variance and the Yaglom limit is in Chapter [Mathematical introduction]; here only (3.1) and (3.5) are needed.

We adopt the standing convention $A(0) := \lim_{p\downarrow 0} A(p) = \frac{1}{2}$ when endpoint statements require it. The continuous-time comparison constant

$$A_{\mathrm{c}}(p) = \frac{1-2p}{1-p} \tag{3.6}$$

was derived in Chapter [Mathematical introduction]from the linear birth–death process with $p = \lambda/(\lambda+\mu)$; that derivation is not repeated. Write $\varepsilon = 1 - 2p$ for the distance from criticality and $r = 2p$ for the logistic multiplier.

3.3 An infinite-product representation

Take the nonlinear map for the survival probability,

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1,$$

and substitute $S_n = 2w_n$. This gives

$$w_{n+1} = 2p\,w_n(1-w_n) = r\,w_n(1-w_n), \qquad r = 2p, \quad w_0 = \tfrac{1}{2}, \tag{3.7}$$

which is the logistic map—the same map that exhibits the period-doubling route to chaos as r increases from 0 to 4 (figure 3.1). The subcritical window is $r \in [0,1)$, at the extreme left of that picture, where the origin is a globally attracting fixed point. That this innocuous corner of the logistic family is nevertheless enough to obstruct a closed form is taken up in section 3.8.

In terms of w, (3.1) reads

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} = 2 \lim_{n\to\infty} \frac{w_n}{r^n}. \tag{3.8}$$

Any sequence can be written as a telescoping product,

$$x_n = x_0 \left(\frac{x_1}{x_0}\right)\left(\frac{x_2}{x_1}\right)\cdots\left(\frac{x_n}{x_{n-1}}\right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k}, \qquad \text{so} \qquad \lim_{n\to\infty} x_n = x_0 \prod_{k=0}^{\infty} \frac{x_{k+1}}{x_k},$$

and applying this with $x_n = w_n/r^n$ the ratio simplifies at once:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{r w_k(1-w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k.$$

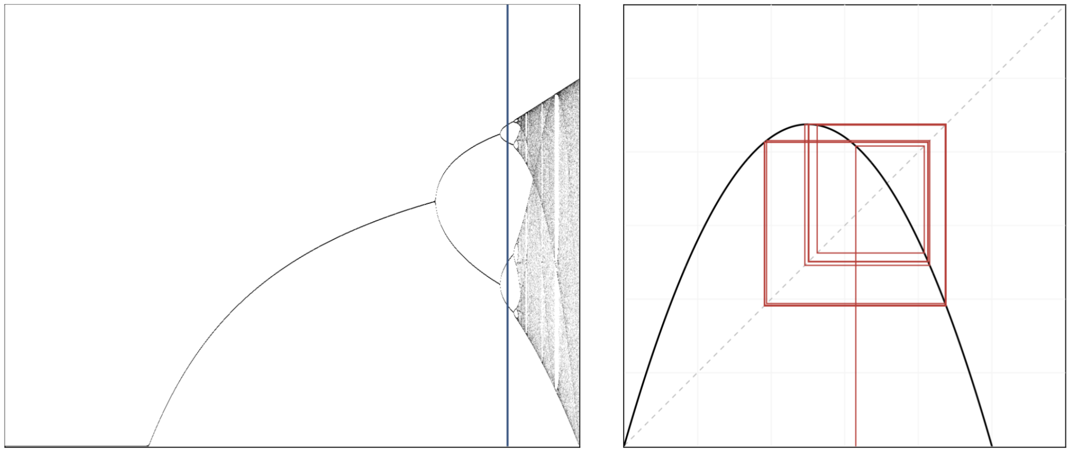

Figure 3.1: Bifurcation diagram of the logistic map $w \mapsto rw(1-w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$; the right-hand plot shows a trajectory of the map at the same r. The regime relevant to subcritical branching is $r = 2p \in [0, 1)$, off the left-hand edge of this diagram, where the dynamics are as simple as they can be.

Hence, with $x_0 = w_0 = \frac{1}{2}$ and $A(p) = 2 \lim_n x_n$,

$$A(p) = 2 \cdot \tfrac{1}{2} \prod_{k=0}^{\infty} (1 - w_k) = \prod_{k=0}^{\infty} (1 - w_k),$$

and pulling out the factor $1 - w_0 = \frac{1}{2}$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \tag{3.9}$$

Convergence

For $0 < w_k < 1$ the infinite product $\prod_{k \geq 1}(1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k \geq 1} w_k < \infty$. From (3.7) with $0 < r < 1$ and $0 < w_n < 1$ we have $w_{n+1} < rw_n$, so by induction $w_n \leq w_0 r^n$; since $0 < r < 1$ this bound is summable, and the product converges to a positive finite value. That also establishes what was asserted in (3.4): the limit (3.1) defining $A(p)$ exists and lies strictly between 0 and $\frac{1}{2}$ for $p \in (0, \frac{1}{2})$.

Is this an advance?

Only a modest one. (3.9) is exact and its partial products decrease monotonically, so truncating gives a rigorous upper bound on $A(p)$ at every order, which the raw ratio $S_n/(2p)^n$ does not. But computing the factors still requires iterating the nonlinear map to generate the S_k, so the near-critical difficulty is untouched. What the product does supply is a route to the sharper result of the next subsection.

3.4 An exact series and two-sided bounds

The product converts into a telescoping sum, and the sum is much more informative. Write $r = 2p$ and note that the recursion factorises as

$$S_{n+1} = rS_n\left(1 - \frac{S_n}{2}\right).$$

Define $v_n = r^n/S_n$, so that $v_n \to 1/A(p)$ by (3.1). Then

$$v_{n+1} - v_n = \frac{r^{n+1}}{rS_n\left(1-\frac{S_n}{2}\right)} - \frac{r^n}{S_n} = \frac{r^n}{S_n}\left[\frac{1}{1-\frac{S_n}{2}} - 1\right] = \frac{r^n}{S_n}\cdot\frac{S_n/2}{1-\frac{S_n}{2}} = \frac{r^n}{2-S_n}.$$

Summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 3.1 (Series representation)**.** *For $p \in [0, \frac{1}{2})$, with S_n generated by (3.3),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \tag{3.10}$$

Unlike the product, this representation isolates the geometric decay in a factor that does not depend on the dynamics at all; the whole of the nonlinearity sits in the bounded denominator $2 - S_n$. That is what makes it useful, because S_n is confined to $(0, 1]$ and therefore $2 - S_n$ to $[1, 2)$. Bounding the denominator by its extremes and summing the resulting geometric series gives the following.

Proposition 3.2 (Two-sided bounds)**.** *With $\varepsilon = 1 - 2p$ as in $\varepsilon = 1 - 2p$, for $p \in (0, \frac{1}{2})$,*

$$\frac{\varepsilon}{1+\varepsilon} \;<\; A(p) \;<\; \frac{2\varepsilon}{1+2\varepsilon}, \tag{3.11}$$

that is,

$$\frac{1-2p}{2(1-p)} \;<\; A(p) \;<\; \frac{2(1-2p)}{3-4p}. \tag{3.12}$$

The lower bound is exactly half the continuous-time constant (3.6):

$$A(p) > \tfrac{1}{2}A_{\mathrm{c}}(p). \tag{3.13}$$

Proof. Since $0 < S_n \le 1$ for all n we have $1 \le 2 - S_n < 2$, and hence $\frac{1}{2}r^n < r^n/(2 - S_n) \le r^n$, with equality on the right only at $n = 0$, where $S_0 = 1$. Summing over $n \ge 0$ and using $\sum_n r^n = 1/(1 - r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} \;<\; \frac{1}{A(p)} \;<\; 1 + \frac{1}{\varepsilon},$$

and inverting gives (3.11). Substituting $\varepsilon = 1-2p$ gives (3.12), whose lower bound is $(1-2p)/(2(1-p)) = \frac{1}{2}A_{\mathrm{c}}(p)$ by (3.6). □

Two features of proposition 3.2 deserve comment, because between them they account for the whole shape of the discrete curve.

At the far subcritical end, $p \to 0$, we have $\varepsilon \to 1$ and the lower bound tends to $\frac{1}{2}$. Since $A(p) \le \frac{1}{2}$ by proposition 3.3 below, the bound is asymptotically attained and $A(0^+) = \frac{1}{2}$. At the critical end, $p \to \frac{1}{2}^-$, we have $\varepsilon \to 0$, the lower bound behaves like ε and the upper like 2ε; section 3.5 shows that it is the upper bound that is attained there. So $A(p)$ runs from its lower bound at one end of the interval to its upper bound at the other, which is why no constant rescaling of A_{c} can reproduce it.

Proposition 3.3 (Parity bound)**.** *For the binary Galton–Watson process started from a single individual, $A(p) \le \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \to 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so Z_n is even for every $n \ge 1$. Conditional on survival, therefore, $Z_n \ge 2$, and hence $\mathbb{E}(Z_n \mid Z_n > 0) \ge 2$ for all $n \ge 1$. Letting $n \to \infty$ and using (3.5) gives $1/A(p) \ge 2$. As $p \to 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit. ☐

Table 3.1: The constant $A(p)$ computed from the product (3.9) in 60-digit arithmetic, against the bounds of proposition 3.2 and the near-critical asymptotic $2(1-2p)$ of section 3.5. The bounds are loose in the middle of the range and tighten at both ends: at $p = 0.05$ the lower bound is already within 3% of A, whereas at $p = 0.4999$ it is the upper bound that is within 0.1%. The last two columns give the quasi-stationary mean and the number of factors of (3.9) needed for the product to reach full precision — the cost that section 3.11 has to manage.

p	$\frac{1}{2}A_c(p)$	$A(p)$	$\dfrac{2(1-2p)}{3-4p}$	$2(1-2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

3.5 Behaviour as $p \to \frac{1}{2}^-$

The series (3.10) also delivers the near-critical asymptotic, and does so more transparently than the matching argument one is first tempted to try.

Fix ε small. The summand $r^n/(2-S_n)$ involves two decays on very different scales. The geometric factor $r^n = (1-\varepsilon)^n$ decays on the scale $n \sim 1/\varepsilon$, which is large. The survival probability S_n, by contrast, is close to its critical behaviour $S_n \sim 2/n$ over that whole range, and so has already fallen to near zero after $O(1)$ generations. For the overwhelming majority of the terms that carry weight, therefore, $S_n \approx 0$ and the summand is approximately $r^n/2$. Summing,

$$\sum_{n=0}^{\infty} \frac{r^n}{2-S_n} \approx \frac{1}{2}\sum_{n=0}^{\infty} r^n = \frac{1}{2\varepsilon},$$

so $1/A(p) \sim 1/(2\varepsilon)$ and

$$\boxed{A(p) \sim 2\varepsilon = 2(1-2p) \qquad \text{as } p \to \tfrac{1}{2}^-.} \tag{3.14}$$

Equivalently $1/A(p) \sim 1/(2(1-2p))$: the quasi-stationary mean diverges like the reciprocal of the distance from criticality. In particular $A(p)$ meets the axis at $p = \frac{1}{2}$ with gradient -4, a fact used repeatedly below as a test of candidate formulae.

Theorem 3.4 (Near-critical survival amplitude)**.** *As $p \uparrow \frac{1}{2}$, with $\varepsilon = 1 - 2p \downarrow 0$,*

$$A(p) = 2\varepsilon \left[1 + O\left(\varepsilon \log \frac{1}{\varepsilon}\right)\right]. \tag{3.15}$$

In particular $A(p) \sim 2(1 - 2p)$ and $1/A(p) \sim 1/(2(1 - 2p))$.

The series argument above yields the leading asymptotic. A fully rigorous bound on the remainder proceeds by decomposing $1/(2 - S_n) = \frac{1}{2} + S_n/(2(2 - S_n))$ and controlling the error series by the critical bound $S_n \leq 2/(n + 2)$, which produces an $O(\log(1/\varepsilon))$ remainder before inversion [2]; the details match the argument recorded in the engineering source for this chapter. The empirical fit $1 - A/(2\varepsilon) \approx \varepsilon(\log(1/\varepsilon) + 0.78)$ is consistent with the order but is not used subsequently.

The same result can be reached through the continuum equation of Chapter [Mathematical introduction], and it is worth seeing why that route is the worse one. The exact difference is

$$S_{n+1} - S_n = (2p - 1)S_n - pS_n^2 = -\varepsilon S_n - \tfrac{1}{2}S_n^2 + \tfrac{\varepsilon}{2}S_n^2,$$

and dropping the $O(\varepsilon S_n^2)$ term gives $\mathrm{d}S/\mathrm{d}n = -\varepsilon S - \frac{1}{2}S^2$. Setting $u = 1/S$ linearises this to $u' = \varepsilon u + \frac{1}{2}$, with solution

$$u(n) = C\mathrm{e}^{\varepsilon n} - \frac{1}{2\varepsilon}, \qquad S(n) = \frac{1}{C\mathrm{e}^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim A\mathrm{e}^{-\varepsilon n}$ for small ε identifies $A = 1/C$, and matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$, recovering (3.14). Both steps are heuristic, and the second is delicate; the series argument is preferable because it needs no second matching at all.

Remark 3.5 (Rate of approach). Numerically, $A(p)/2\varepsilon$ approaches 1 rather slowly. Evaluating (3.9) in high precision gives $A/2\varepsilon = 0.90987$, 0.98612, 0.99814, 0.99977 and 0.999972 at $\varepsilon = 2 \times 10^{-2}$ down to 2×10^{-6}. Fitting the deficit gives $1 - A/2\varepsilon \approx \varepsilon\big(\ln(1/\varepsilon) + c\big)$ with $c \approx 0.78$, so

$$A(p) = 2\varepsilon\Big[1 + O\big(\varepsilon \ln(1/\varepsilon)\big)\Big].$$

The logarithmic correction is why (3.14) is a good practical surrogate only very close to criticality, and it is what fixes the crossover point chosen in section 3.11.

3.6 Why the discrete and continuous constants differ

The question raised by figure 3.2 can now be settled. The two constants are

$$A_{\mathrm{c}}(p) = \frac{1 - 2p}{1 - p} \quad \text{(continuous time)}, \qquad A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} \quad \text{(discrete time)},$$

and the observation was that the first starts at 1 where the second starts at $\frac{1}{2}$, yet halving the first does not superpose the curves.

Proposition 3.2 explains this completely. What holds is not $A = \frac{1}{2}A_{\mathrm{c}}$ but the strict inequality $A > \frac{1}{2}A_{\mathrm{c}}$, and the inequality is attained only in the limit $p \to 0$. Combining (3.6) and (3.14), the ratio runs monotonically across the interval:

$$\frac{A(p)}{A_{\mathrm{c}}(p)} \longrightarrow \tfrac{1}{2} \quad (p \to 0), \qquad \frac{A(p)}{A_{\mathrm{c}}(p)} \longrightarrow 1 \quad (p \to \tfrac{1}{2}^{-}), \tag{3.16}$$

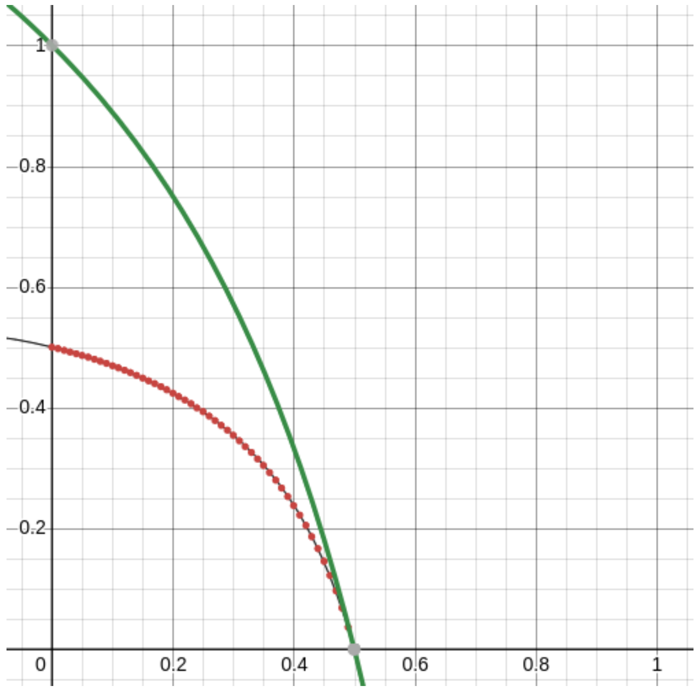

Figure 3.2: The continuous-time constant $A_c(p) = (1-2p)/(1-p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$; at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$.

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$. No constant factor will do, because the ratio is not constant.

The two endpoints have clean interpretations, and they are different in kind.

At $p \to 0$ the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so a surviving population is always even (proposition 3.3); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model a survivor is a single individual that has not yet died, so the quasi-stationary mean is 1. The reciprocals are $\frac{1}{2}$ and 1, and there is the discrepancy.

At $p \to \frac{1}{2}^-$ the factor disappears, because both constants are governed by the same critical scaling. Expanding (3.6) near $p = \frac{1}{2}$ gives $A_c(p) = (1-2p)/(1-p) \to 2(1-2p)$, which is precisely (3.14). Near criticality the discreteness of the generational clock stops mattering; what survives is the universal 2ε behaviour, and the two models agree to leading order.

The discrete curve is therefore squeezed between $\frac{1}{2}A_c$ and its own critical asymptote, hugging the former at one end and the latter at the other. That is the whole content of figure 3.2.

3.7 Searching for a closed form

Before the connection of section 3.8 was appreciated, several methods were tried in the hope of an elementary formula for $A(p)$. The negative result is worth recording, and so is the manner of the failure.

The first method was the continuum route of Chapter [Mathematical introduction]: treat the map as a differential equation in the large-generation limit and solve it. This gives the shape of the decay but leaves the constant of integration undetermined, which is to say it leaves $A(p)$ undetermined.

The second was symbolic regression by genetic algorithm. The idea is simple: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those; then measure again, and repeat the cycle of variation and selection many times over [3, 4]. The first implementation searched a space of polynomials, on the hunch that the answer might take that form and because such functions are simple to encode: a genome is a coefficient list $[a_0, \ldots, a_k]$ for $a_0 + a_1 p + \cdots + a_k p^k$ up to some maximum order, drawn uniformly at random. It became apparent quickly that many substantially different formulae produce curves indistinguishable from the target by eye. Supposing that a true formula might have rational coefficients, a second version searched $\sum_i (a_i/b_i)p^i$ with $[a_i], [b_i] \in \mathbb{Z}^{k+1}$, and a third extended the space to ratios of two such polynomials. The best that emerged merely fitted the high-precision numerical data well:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \tag{3.17}$$

This passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$ and looks convincing, but it fails the diagnostic of (3.14): differentiating at $p = \frac{1}{2}$ gives a gradient of $-2/0.55 = -3.636$ rather than the required -4.

A later attempt used a symbolic tree search over progressively nested elementary functions of all kinds. The conclusion was the same. Many functions of widely varying form resemble the target closely, and no satisfactory simple formula emerges even when candidates with fewer components are explicitly favoured. The expressions returned were typically deep nestings of trigonometric and iterated exponential terms carrying a dozen or more fitted decimal constants, and they were not interpretable in any useful sense; a representative and comparatively concise one is

$$\hat{A}_3(p) = \frac{-0.616647881739505\,\mathrm{e}^{-0.552605335919693\,\mathrm{e}^{p}}}{\sin(\cos p) - |p|} + 0.921964323679771. \tag{3.18}$$

The recurring defect is the same in every case. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4, and near criticality this matters enormously: as $A \to 0$, an error in A of vanishing absolute size produces an $O(1)$ — eventually unbounded — error in the quasi-stationary mean $1/A$. A formula can look perfect on a plot of A and be useless on a plot of $1/A$.

A pragmatic patch is to splice the asymptotic onto a fitted curve,

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545\,\mathrm{e}^{\mathrm{e}^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962, \end{cases} \tag{3.19}$$

and this does work. But if one is going to splice, it is better to splice onto something exact, and section 3.11 does so.

The deeper lesson points directly at the next subsection. That an enormous number of very different formulae can behave almost identically over a bounded interval is not merely an inconvenience of the method; it is a symptom. Curve fitting cannot distinguish a function that has a closed form from one that does not.

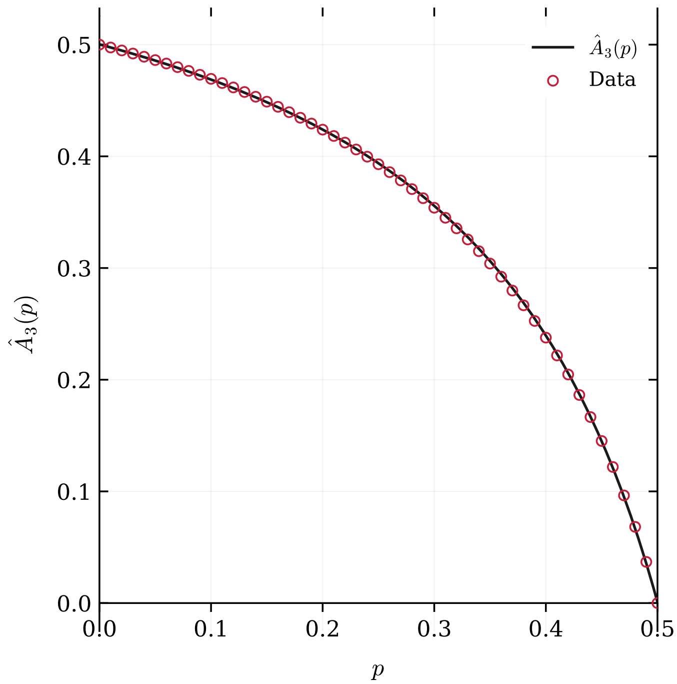

Figure 3.3: The symbolic-regression candidate $\hat{A}_3(p)$ of (3.18) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point, returning $\hat{A}_3(\frac{1}{2}) = 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient -3.63 instead of -4. Errors of this size are invisible here but are fatal for $1/A$ near criticality.

3.8 The Koenigs connection

Some time after the search had been abandoned, a dynamical argument emerged that explains why no elementary closed form should be expected. In short: $A(p)$ is a value of the Koenigs linearising function of the logistic map, and that function is known to be non-elementary throughout the parameter range required.

Definition 3.6 (Koenigs function)**.** Let $f(z) = rz(1-z)$ have a fixed point at $z = 0$ with multiplier $r = f'(0)$, and suppose $|r| \notin \{0, 1\}$. Koenigs [5] proved that there is a unique function ψ_r, analytic near the origin, satisfying the Schröder equation

$$\psi_r\big(f(z)\big) = r\,\psi_r(z), \qquad \psi_r(0) = 0, \qquad \psi_r'(0) = 1. \tag{3.20}$$

The Koenigs function is a change of coordinate that straightens the dynamics: in the ψ coordinate the nonlinear map f becomes multiplication by r (figure 3.4). Here $f(z) = rz(1-z)$ is exactly (3.7), with $z = w_n$ and $f(z) = w_{n+1}$, and the multiplier is $r = 2p \in (0, 1)$ — an attracting fixed point, which is the classical Koenigs setting. Standard accounts are Milnor [6] and Carleson and Gamelin [7].

Proposition 3.7. *For $p \in (0, \frac{1}{2})$ and $r = 2p$,*

$$A(p) = 2\,\psi_r\Big(\tfrac{1}{2}\Big)\,. \tag{3.21}$$

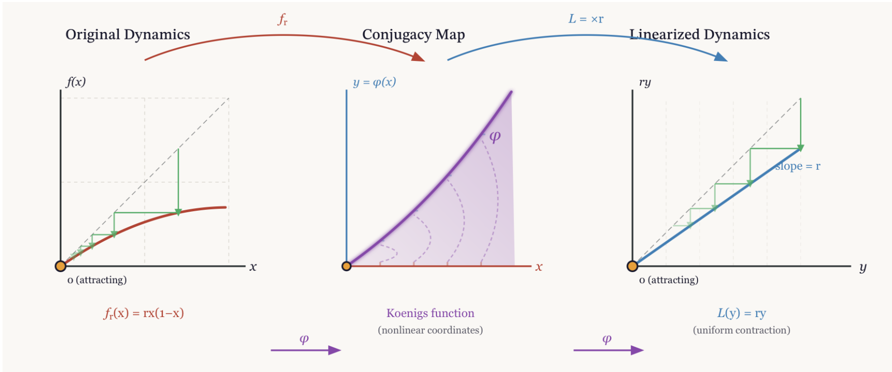

Figure 3.4: Koenigs linearisation of the logistic map near an attracting fixed point. The nonlinear dynamics $x \mapsto f_r(x) = rx(1-x)$ are conjugated by ψ_r to the linear map $y \mapsto ry$; the diagram displays the functional relation $\psi_r(f_r(x)) = r\psi_r(x)$ and the role of ψ_r as the coordinate change that straightens the dynamics.

Proof. Iterating (3.20) along the orbit $w_0, w_1, w_2, \ldots$ of (3.7) gives $\psi(w_1) = r\psi(w_0)$, hence $\psi(w_0) = \psi(w_1)/r$; similarly $\psi(w_1) = \psi(w_2)/r$, so $\psi(w_0) = \psi(w_2)/r^2$; and by induction

$$\psi(w_0) = \frac{\psi(w_n)}{r^n} \qquad \text{for every } n. \tag{3.22}$$

Since (3.22) holds for all n we may pass to the limit, $\psi(w_0) = \lim_n \psi(w_n)/r^n$. The normalisation $\psi(0) = 0$, $\psi'(0) = 1$ means $\psi(z)/z \to 1$ as $z \to 0$, and $w_n = S_n/2 \to 0$ because $r < 1$; replacing $\psi(w_n)$ by w_n in the limit is therefore legitimate, and $\psi(w_0) = \lim_n w_n/r^n$. Comparing with (3.8) and using $w_0 = S_0/2 = \frac{1}{2}$ gives $A(p) = 2\psi_r(\frac{1}{2})$. □

Elementary cases and the obstruction

At the exceptional parameters $r = 2$ and $r = 4$ — both *outside* the branching window $r = 2p \in (0, 1)$ — the linearisations are elementary:

$$\psi_2(x) = -\tfrac{1}{2}\ln(1-2x), \qquad \psi_4(x) = \left(\arcsin\sqrt{x}\right)^2. \tag{3.23}$$

Both sit over *repelling* fixed points and are derived in full in appendix .1. For $r \in (0, 1)$ the fixed point is attracting, and the precise hypertranscendence theorem is stated in section 3.9: for every such r, ψ_r is hypertranscendental over $\mathbb{C}(z)$. The softer parameter-plane observation that $c(r) = (2r - r^2)/4$ avoids $\{0, -2\}$ on $r \in (0, 1)$ is true but logically superfluous once the attracting/repelling dichotomy is in hand (remark 3.14).

3.9 Hypertranscendence of the conjugacy

This section makes the obstruction precise and delimits exactly what it does and does not assert. The logical skeleton is short. First, every elementary function satisfies an algebraic differential equation. Second, differential algebraicity survives compositional inversion. Third, by a theorem of Becker and Bergweiler, a solution of Schröder's equation for a rational map of degree at least two can be differentially algebraic only when the underlying fixed point is *repelling*. Fourth, for $r \in (0, 1)$ the fixed point 0 of f_r is attracting. Hypertranscendence of ψ_r for every subcritical parameter

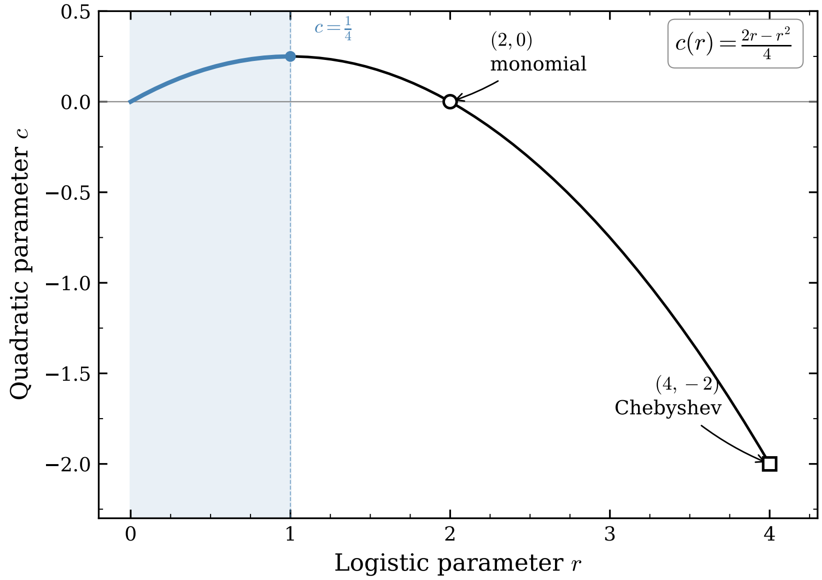

Figure 3.5: The affine parameter correspondence $c(r) = (2r - r^2)/4$ between the logistic family $f_r(x) = rx(1-x)$ and the quadratic family $P_c(z) = z^2 + c$. The shaded segment is $r \in (0, 1)$, the subcritical branching window, which maps to $c \in (0, \frac{1}{4})$. The two integrable parameters, $r = 2$ ($c = 0$, monomial) and $r = 4$ ($c = -2$, Chebyshev), lie outside it.

follows at once—with no exceptional cases to exclude, because the attracting window contains no candidates at all. We then explain, with equal care, why this theorem does *not* by itself settle the status of the individual numbers $A(p_0)$ or of the map $p \mapsto A(p)$, and we record the numerical evidence bearing on those.

3.9.1 Differential algebraicity, hypertranscendence, elementarity

Definition 3.8 (differentially algebraic; hypertranscendental)**.** A function g, holomorphic on a domain (or a formal power series), is *differentially algebraic over* $\mathbb{C}(z)$ if there exists a non-zero polynomial P in $n+2$ variables, for some $n \geq 0$, with

$$P\big(z,\, g(z),\, g'(z),\, \ldots,\, g^{(n)}(z)\big) \equiv 0.$$

Otherwise g is *differentially transcendental*, or *hypertranscendental*.

Two closure facts make this the right notion for our purpose.

Lemma 3.9 (elementary $\Rightarrow$ differentially algebraic)**.** *Every elementary function in the Liouville sense — any function reachable from $\mathbb{C}(z)$ by a finite tower of algebraic operations, exponentials and logarithms — is differentially algebraic. Consequently a hypertranscendental function is not elementary.*

Proof sketch. Each storey of a Liouvillian tower preserves differential algebraicity: algebraic extensions do so by elimination; if g is differentially algebraic then so are e^g and $\log g$, their defining first-order relations combining with the equation for g; and the class is closed under field operations and composition. These closure properties are classical; see Rubel's survey [8]. $\square$

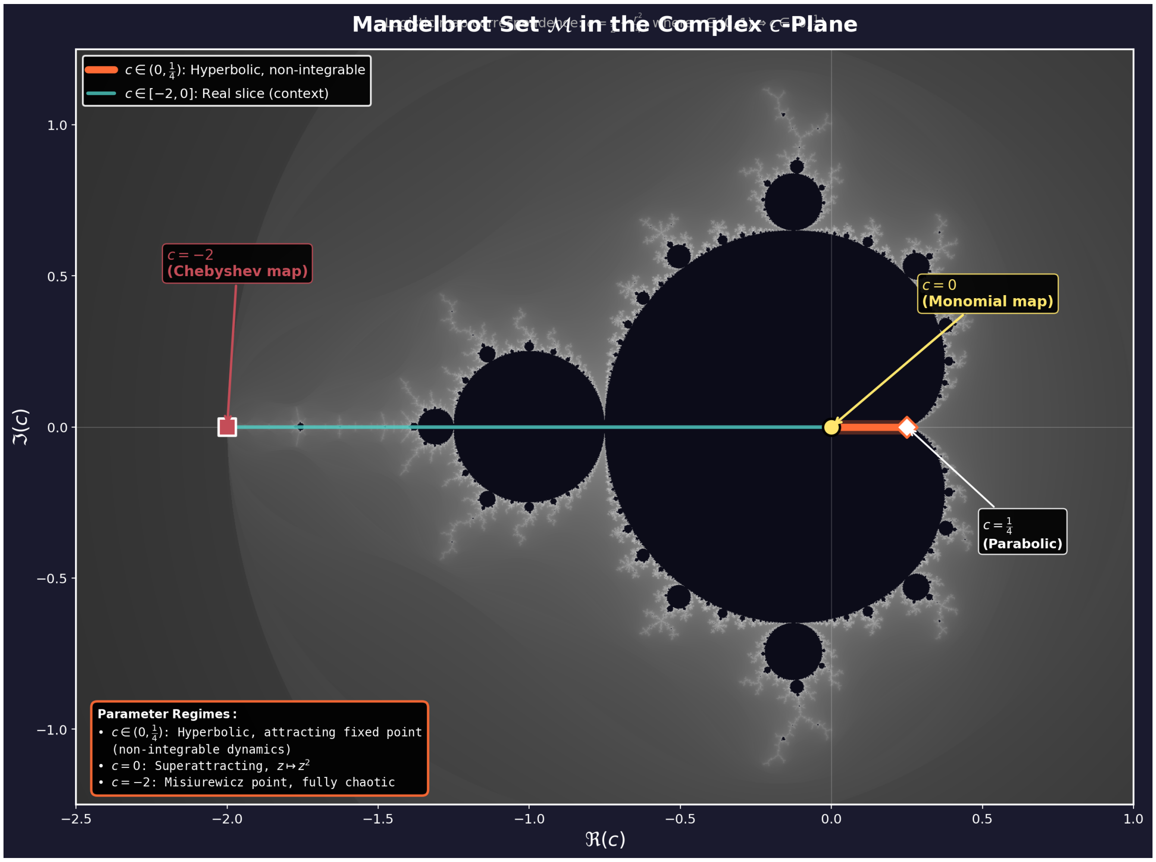

Figure 3.6: The Mandelbrot set in the complex c-plane, with the interval $c \in (0, \frac{1}{4})$ highlighted. Under the correspondence $c = (2r - r^2)/4$ this is the image of the subcritical branching window $r = 2p \in (0, 1)$. Pedagogical context only: the hypertranscendence proof of section 3.9 does not rely on this picture.

Lemma 3.10 (closure under compositional inversion)**.** *Let g be holomorphic near 0 with $g(0) = 0$ and $g'(0) \neq 0$, and let g^{-1} denote its local compositional inverse. Then g is differentially algebraic over $\mathbb{C}(z)$ if and only if g^{-1} is (Moore [9]; see also Boshernitzan–Rubel [10]).*

Lemma 3.10 matters because the literature states Schröder-equation results in two equivalent but opposite conventions, and conflating them is an easy way to misquote a theorem. Our ψ_r is the *linearising coordinate*: $\psi_r(f_r(z)) = r\,\psi_r(z)$. The function classified by Becker and Bergweiler (and by Ritt before them) is the *linearising parametrisation* $\varphi_r = \psi_r^{-1}$, which satisfies

$$\varphi_r(r\,t) = f_r(\varphi_r(t)), \qquad \varphi_r(0) = 0, \quad \varphi_r'(0) = 1. \tag{3.24}$$

By Lemma 3.10, hypertranscendence of either function transfers to the other.

3.9.2 The Becker–Bergweiler theorem, stated precisely

The result we need is the 1995 classification of differentially algebraic Schröder functions [1]; we quote it in the form given in Fernandes' survey of the Schröder, Böttcher and Abel equations [11, Thm 3.1].

Theorem 3.11 (Becker–Bergweiler [1]; as restated in [11, Thm 3.1])**.** *Let R be a rational map of degree at least two with $R(0) = 0$ and multiplier $s = R'(0)$, where $s \neq 0$ and s is not a root of unity (for $|s| = 1$ the Schröder function is understood as a formal series). Let φ, normalised by $\varphi(0) = 0$ and $\varphi'(0) = 1$, solve $\varphi(st) = R(\varphi(t))$. If φ is differentially algebraic over $\mathbb{C}(t)$, then:*

(i) *the fixed point* 0 *is* repelling, $|s| > 1$*; and*

(ii) φ *is a Möbius transformation of one of*

$$\exp(\alpha t^\rho), \qquad \cos(\alpha t^\rho + \beta), \qquad \wp(\alpha t^\rho + \beta),\ \wp^2,\ \wp^3,\ \wp'(\alpha t^\rho + \beta),$$

where $\wp$ is a Weierstrass function, $\rho \in \mathbb{Q}$ is such that the function concerned is meromorphic on $\mathbb{C}$, $\alpha \neq 0$, and β is a fraction of a period. Correspondingly R is Möbius-conjugate to $z^{\pm d}$, to $\pm T_d$ (Chebyshev), or to a Lattès map.

Remark 3.12 (lineage and independent corroboration). For $|s| > 1$ — the Poincaré-function case — the classification goes back to Ritt in 1926 [12]; Becker and Bergweiler completed the picture for the three classical functional equations at once, and Bergweiler's later account with Aschenbrenner [13] assigns the exceptional list explicitly to the repelling case. Di Vizio and Fernandes [14] have given an independent difference-Galois proof of Ritt's theorem; notably, their Theorems 1.2 and 4.2 assume only that s is neither zero nor a root of unity — no condition $|s| > 1$ — and show that any differentially algebraic solution of $y(R(x)) = s\,y(x)$ (our ψ-convention, with no inversion needed) must already satisfy a first-order equation of Ritt type $(y')^\rho = B(x)\,y^j$. The structure underlying Theorem 3.11 therefore does not hang on a single paper.

3.9.3 Hypertranscendence throughout the subcritical window

Theorem 3.13 (no exceptional parameters in $(0, 1)$)**.** *For every $r \in (0, 1)$, the Koenigs function ψ_r of $f_r(z) = rz(1 - z)$ at 0 is hypertranscendental over $\mathbb{C}(z)$: it satisfies no algebraic differential equation. In particular, ψ_r is not an elementary function.*

Proof. Fix $r \in (0, 1)$. The map f_r is rational of degree two, fixes 0, and its multiplier $s = f_r'(0) = r$ satisfies $0 < s < 1$; the hypotheses of Theorem 3.11 hold, with 0 attracting. Suppose, for contradiction, that ψ_r were differentially algebraic. Its local inverse $\varphi_r = \psi_r^{-1}$ solves the parametrisation form (3.24) with the stated normalisation, and by Lemma 3.10 it would be differentially algebraic too. Theorem 3.11(i) would then force $|r| > 1$, contradicting $r \in (0, 1)$. Hence ψ_r is hypertranscendental, and non-elementarity follows from Lemma 3.9. □

Remark 3.14 (the exceptional set inside the window is empty, not merely avoided). An earlier form of this argument reasoned through the parameter plane: the affine conjugacy $c(r) = (2r - r^2)/4$ of Lemma .16 sends $r \in (0, 1)$ to $c \in (0, \frac{1}{4})$, which avoids the exceptional quadratic parameters $c = 0$ (reached at $r = 2$) and $c = -2$ (reached at $r = 4$, and also at $r = -2$). That observation is true but logically superfluous, and it invites a question it cannot answer — *is the exceptional list, restricted to the window, complete?* The route through Theorem 3.11(i) dissolves the question: the exceptional Schröder functions exist only over *repelling* fixed points, so the attracting window contains no candidates whatsoever, and no enumeration of exceptional parameters is required. Consistently, the classical elementary cases of appendix .1 have multipliers 2 and 4 at the linearised fixed point — both repelling: they are Poincaré-type linearisations, and neither can be pressed into service for $r \in (0, 1)$.

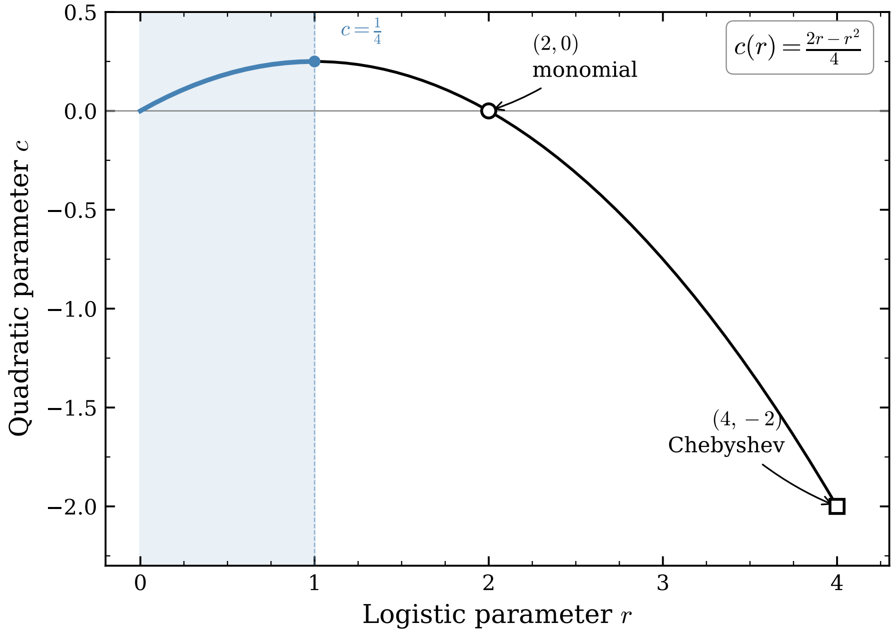

Figure 3.7: Parameter correspondence $c(r) = (2r - r^2)/4$. The segment $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$, disjoint from the integrable values $c \in \{0, -2\}$. In the light of Remark 3.14 this picture is a consistency check rather than the proof: the exceptional linearisations live over repelling fixed points, which the attracting window excludes outright.

Remark 3.15 (germ versus basin). Theorem 3.13 concerns ψ_r as an analytic function; the quantity $A(p) = 2\psi_r(\frac{1}{2})$ of (3.21) uses its extension to the real basin of attraction, since for $p \geq \frac{1}{3}$ the point $\frac{1}{2}$ lies outside the disc on which Koenigs' series is guaranteed to converge (radius $(1 - r)/r$). The extension is by finitely many applications of the functional equation,

$$\psi_r(w) = r^{-N}\, \psi_r\big(f_r^{\circ N}(w)\big) \quad \text{for } N \text{ large enough that } f_r^{\circ N}(w) \text{ enters the germ,}$$

and an algebraic differential equation, if one held on any subdomain, would persist under this analytic continuation. Hypertranscendence of the germ and of the basin extension are therefore equivalent statements, and the identity (3.21) is unaffected.

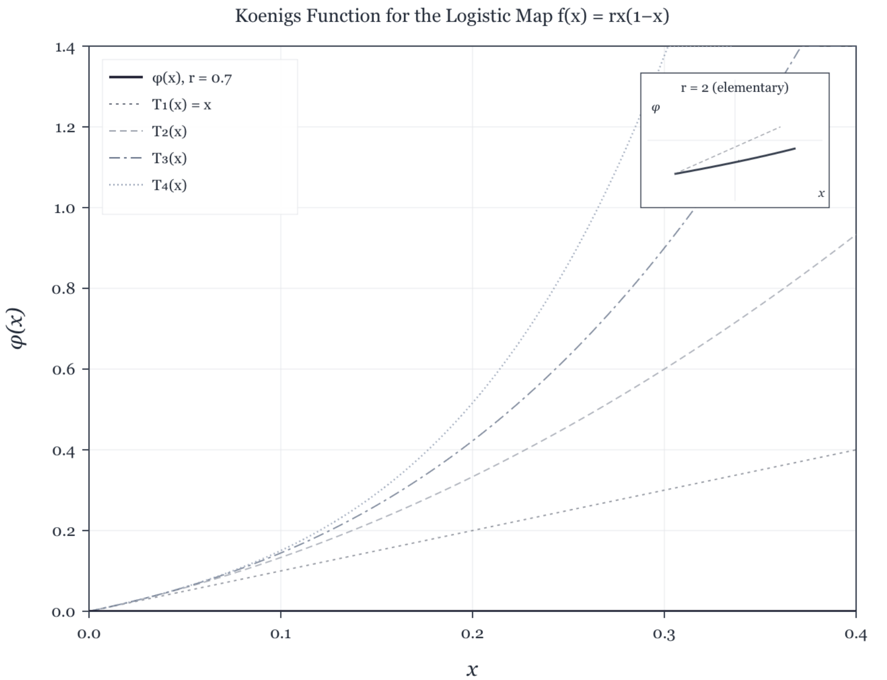

Figure 3.8: Numerical approximation of a Koenigs function for a logistic multiplier in $(0, 1)$, with low-order Taylor polynomials near the origin. Elementary closed forms exist only for exceptional multipliers outside this range (appendix .1).

3.10 Scope: what is and is not proved about $A(p)$

Before any claim about closed forms is stated, the phrase itself must be made precise: at least four inequivalent readings are in play, and a result for one is not a result for another.

(V) the *value* $A(p_0)$ is elementary or algebraic, for a fixed rational p_0;

(E) the *map* $p \mapsto A(p)$ is an elementary function of p;

(D) the map is *differentially algebraic in* p, i.e. satisfies some algebraic ODE in p;

(S) the map is expressible via standard special functions (Γ, ζ, ${}_2F_1$, theta or q-series).

Theorem 3.13 speaks to the function $z \mapsto \psi_r(z)$ at fixed r, and licenses exactly this much: *no evaluation of* $A(p)$ *can be short-circuited through an algebraic differential equation for the lineariser, at any subcritical parameter.* It does not, by itself, decide any of (V), (E), (D) or (S). Two gaps must be stated plainly.

The value gap. A hypertranscendental function can take elementary — even algebraic — values at individual points: Γ is hypertranscendental by Hölder's theorem, yet $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Nothing so far forbids, say, $A(\frac{1}{4})$ from being an algebraic number. There *is* a genuine values theorem in this circle of ideas: Becker and Bergweiler proved in 1993 [15] that the Böttcher conjugacy between two polynomials of the *same degree* $d \geq 2$ with algebraic coefficients (not linearly conjugate to monomials or Chebyshev polynomials) is transcendental and takes transcendental values at algebraic points; Mahler's method drives the proof (for the method, see Nishioka [16]). One might hope to transfer this to the Koenigs value $\psi_r(\frac{1}{2})$. It does not transfer, for two independent reasons. *Formally*, the Schröder equation pairs an inner map of degree two with an outer map — multiplication by r — of degree one, so the equal-degree hypothesis fails outright. *Structurally*, no easy repair exists: Mahler's method balances doubly-exponential analytic decay along an orbit against comparable height growth, and in the Böttcher setting both exponents run like d^n; in the attracting Koenigs setting the orbit $w_n = f_r^{\circ n}(\frac{1}{2})$ decays only geometrically, $|w_n| \sim A\,r^n$, while the heights of the rational numbers w_n still grow doubly exponentially, $h(w_n) \sim C\,2^n$, and the resulting inequalities run the wrong way at every n. Note that the obstruction is *not* the germ/basin issue — the pullback of Remark 3.15 moves the evaluation point arbitrarily deep into the germ at the cost of a rational factor — it is the degree structure of the equation itself. The consequence deserves emphasis:

transcendence — indeed, even irrationality — of $A(p_0)$ is at present not proved for a single rational $p_0 \in (0, \frac{1}{2})$.

The parameter gap. Theorem 3.13 says nothing about the map $p \mapsto A(p)$: readings (E) and (D) are open. To the best of our knowledge the question is not merely unsolved but unaddressed in the literature. The natural machinery — the parametrised differential Galois theory of difference equations of Hardouin and Singer [17], designed precisely to detect differential relations in auxiliary directions for linear difference equations such as $\sigma\psi = r\psi$ — does not apply as it stands, because the derivation ∂_r fails to commute with the substitution operator $\sigma\colon w \mapsto f_r(w)$, which itself moves with r. Nor does any feature of the near-critical asymptotics obstruct (E) or (D): logarithmic corrections to (3.14) are themselves elementary functions of ε.

3.10.1 Inverse-symbolic evidence at eleven rational parameters

Since the theorems leave (V) open, the value question was attacked directly by integer-relation detection. The values

$$A(p), \qquad p \in \left\{\tfrac{1}{10}, \tfrac{1}{8}, \tfrac{1}{6}, \tfrac{1}{5}, \tfrac{1}{4}, \tfrac{3}{10}, \tfrac{1}{3}, \tfrac{3}{8}, \tfrac{2}{5}, \tfrac{5}{12}, \tfrac{9}{20}\right\},$$

were computed to more than 1150 significant digits by two mechanically independent routes — the exact product identity (3.9) iterated to convergence, and a pullback of the orbit into the Koenigs germ followed by evaluation of the Koenigs series — which agreed to at least 1192 digits at every parameter. Each value was then subjected to a PSLQ battery at ≥ 200-digit effective rigour:

- *algebraicity*: no integer polynomial relation of degree ≤ 10 and coefficient height $\leq 10^6$;
- *linearity*: no rational linear relation, at height $\leq 10^8$, against the constants π, π^2, $\ln 2$, $\ln 3$, $\ln 5$, γ, $\zeta(3)$, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, e, e^π, $\Gamma(\frac{1}{3})$, $\Gamma(\frac{1}{4})$ and Catalan's constant (also tested for $1/A$, the quasi-stationary mean itself, and for A/ε);
- *bilinearity*: no representation $A = L_1/L_2$ with L_1, L_2 integer linear forms of height $\leq 10^8$ in a twelve-constant basis (run at 460 digits);
- *multiplicativity*: no relation $A = e^{q_0} 2^{q_1} 3^{q_2} 5^{q_3} 7^{q_4} \pi^{q_5}$ with rational exponents of height $\leq 10^{10}$;
- *cross-parameter*: no rational linear or multiplicative relation linking the eleven values to one another (run at 500 digits).

In all, 123 tests returned no relation — indeed no discovery-stage candidate: every PSLQ run terminated on its internal norm bound, so each null carries a genuine "no relation up to the stated height" guarantee. Throughout, the precision head-room (digits divided by basis size) exceeded every height cap by at least eight orders of magnitude, so a returned relation would have been meaningful rather than numerical noise.[1]

This is finite evidence, and its scope must be stated honestly: it excludes low-complexity closed forms over the tested constants, and nothing more. A closed form over untested constants ($\Gamma(\frac{1}{7})$, say, or periods of higher-genus curves), or one of height above the caps, would have escaped the search. It is evidence for the conjecture, not a proof.

3.10.2 The working claim

> *For every fixed $r = 2p \in (0, 1)$, the Koenigs linearisation ψ_r is hypertranscendental in its own variable (Theorem 3.13) — a theorem, with no exceptional parameters in the attracting window, since differentially algebraic Schröder functions exist only over repelling fixed points. Consequently no elementary expression in z for the conjugacy can be evaluated to produce $A(p)$. Whether individual values $A(p_0)$ are transcendental (or even irrational), and whether the map $p \mapsto A(p)$ is elementary or differentially algebraic in p, remain open questions; no closed form under any reading is known, none is expected, and eleven rational values resist inverse-symbolic identification against standard constants at over* 200*-digit rigour.*

Full derivations of the elementary solutions at $r = 2$ and $r = 4$, and their place inside the Becker–Bergweiler classification, are collected in appendix .1.

3.11 Computing $A(p)$ in practice

In the absence of an exact formula we are left with approximation, and the two regimes of section 3.4 suggest how to split the work.

Over most of the interval, direct iteration is entirely satisfactory: the infinite product (3.9) converges quickly, its partial products bound $A(p)$ from above at every order, and modest numbers of iterations give many digits. The difficulty is confined to a neighbourhood of $p = \frac{1}{2}$, and it has two parts. First, as $A(p) \to 0$ the quantity of interest $1/A$ diverges, so a fixed absolute error in A becomes an unbounded error in the quasi-stationary mean. Second, and this is the binding constraint, as the process is defined ever closer to criticality its lifetime, though still finite, grows enormously, and S_n recedes to zero ever more slowly. The iteration must be carried to a generation at which S_n is essentially unchanging, and table 3.1 shows what that costs: about 10^2 iterations at $p = 0.2$, 10^3 at $p = 0.45$, but nearly 5×10^5 at $p = 0.4999$.

The remedy is to hand the near-critical region to the asymptotic (3.14), which is exact in the limit and cheap everywhere:

$$A(p) \approx \begin{cases} \dfrac{1}{2}\displaystyle\prod_{k=1}^{N}\left(1 - \frac{S_k}{2}\right), & p < p_*, \quad N \gg 1, \\[2ex] 2 - 4p, & p \geq p_*, \end{cases} \tag{3.25}$$

[1] Computations in `mpmath` at working precisions of 310–1200 digits; scripts, the 1150-digit values, and the full per-test log are archived with the project sources.

with the crossover p_* chosen just below $\frac{1}{2}$. Remark 3.5 indicates where to put it. The relative error of the asymptotic is about $\varepsilon \ln(1/\varepsilon)$, so at $p_* = 0.4999$ ($\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate to about two parts in 10^3, while the product still converges in a few hundred thousand iterations. Any p_* in the range 0.499 to 0.4999 gives a scheme accurate to better than one per cent throughout, and the two branches can be blended over a narrow window if continuity of the derivative matters.

This is the recipe used for the numerical values quoted throughout this chapter, including those in the conditional-mean plot of Chapter [Mathematical introduction]and table 3.1.

3.12 Conclusion

The constant $A(p)$ organises the quasi-stationary mean of subcritical binary Galton–Watson branching. It is well defined, computable by the hybrid scheme of section 3.11, and identified with the basin Koenigs value $A(p) = 2\psi_{2p}(\frac{1}{2})$ (proposition 3.7). For every $r \in (0, 1)$ that conjugacy is hypertranscendental in its own variable (theorem 3.13)—a theorem with no exceptional parameters in the attracting window, since differentially algebraic Schröder functions exist only over repelling fixed points. That fact explains the failure of every elementary closed-form search, while leaving $A(p)$ fully usable for analysis and numerics.

What the theorem does not decide is flagged with equal care (section 3.10). Transcendence—indeed, even irrationality—of individual values $A(p_0)$, and the elementarity or differential algebraicity of the map $p \mapsto A(p)$ in p, remain open: the former because the 1993 Becker–Bergweiler values theorem is confined to equal-degree Böttcher pairs and does not transfer; the latter because the question appears nowhere in the literature. Eleven rational values nonetheless resist all inverse-symbolic identification at over 200-digit rigour (section 3.10.1). The exceptional elementary linearisations at $r = 2$ and $r = 4$ (both repelling) lie outside the branching window; their derivations, and their place inside the classification, appear in appendix .1.

For the modelling chapters that follow, the usable conclusions are the product and series formulae, the two-sided bounds $A > \frac{1}{2}A_c$, the near-critical asymptotic with logarithmic remainder, and the practical hybrid evaluation scheme.

.1 Elementary Koenigs functions at $r = 2$ and $r = 4$

These cases lie outside subcritical branching ($r = 2p < 1$) but complete the classical picture of integrable logistic linearisations. The derivations follow the closed-form note integrated into this document.

.1.1 Case $r = 2$ (logarithmic)

For $r = 2$, $f_2(x) = 2x(1 - x)$. The fixed point 0 has multiplier $\lambda = 2$ (repelling); the linearisation is of Poincaré type rather than an attracting-Koenigs function, but an elementary closed form still exists.

The affine change $u = 1 - 2x$ conjugates the dynamics to $u \mapsto u^2$. Linearising $u \mapsto u^2$ by the logarithm and returning to x yields the candidate

$$\psi_2(x) = -\frac{1}{2}\ln(1 - 2x), \tag{26}$$

normalised so that $\psi_2(0) = 0$ and $\psi_2'(0) = 1$.

Verification.

$$\psi_2(f_2(x)) = -\frac{1}{2}\ln(1-2\cdot 2x(1-x)) = -\frac{1}{2}\ln((1-2x)^2) = -\ln(1-2x) = 2\psi_2(x).$$

.1.2 Case $r = 4$ (inverse trigonometric)

For $r = 4$, $f_4(x) = 4x(1-x)$. The fixed point 0 has multiplier $f_4'(0) = 4$ (repelling); as with $r = 2$, this is a Poincaré-type linearisation, not an attracting Koenigs function. With $x = \sin^2\theta$ one finds $f_4(x) = \sin^2(2\theta)$, so the angle doubles. The candidate

$$\psi_4(x) = \left(\arcsin\sqrt{x}\right)^2 \tag{27}$$

satisfies $\psi_4(f_4(x)) = 4\psi_4(x)$ on a suitable real domain, and $\psi_4(x) \sim x$ as $x \to 0$, so $\psi_4'(0) = 1$.

Verification. Using $\arcsin(2u\sqrt{1-u^2}) = 2\arcsin u$ with $u = \sqrt{x}$ (on an appropriate interval),

$$\psi_4(f_4(x)) = \left(\arcsin\sqrt{4x(1-x)}\right)^2 = \left(2\arcsin\sqrt{x}\right)^2 = 4\psi_4(x).$$

.1.3 Affine conjugacy to $z^2 + c$

Lemma .16. *The logistic map $f_r(x) = rx(1-x)$ is affinely conjugate to $P_c(z) = z^2 + c$ with $c = (2r - r^2)/4$.*

Proof sketch. Seek $h(x) = ax + b$ with $h \circ f_r \circ h^{-1} = P_c$. The substitution $x = -z/r + \frac{1}{2}$ (for $r \neq 0$) yields a monic quadratic; matching coefficients gives $c = (2r - r^2)/4$. □

Under this conjugacy, $r = 2 \Leftrightarrow c = 0$ (monomial z^2) and $r = 4 \Leftrightarrow c = -2$ (Chebyshev type; the value $c = -2$ is also reached at $r = -2$). Neither exceptional value lies in the image of $r \in (0,1)$, which is $c \in \left(0, \frac{1}{4}\right)$. As explained in remark 3.14, this parameter count is a consistency check rather than the proof: the decisive point is that both exceptional cases sit over *repelling* fixed points, which the attracting window excludes structurally.

.1.4 The worked examples inside the Becker–Bergweiler classification

The precise classification is theorem 3.11 in the main text: a differentially algebraic Schröder function exists only over a repelling fixed point, and is then a Möbius transformation of $\exp(\alpha t^\rho)$, of $\cos(\alpha t^\rho + \beta)$, or of Weierstrass-$\wp$ material, with the underlying map Möbius-conjugate to $z^{\pm d}$, $\pm T_d$, or a Lattès map. The two derivations above realise the first two families exactly.

Case $r = 2$. Inverting (26): $t = -\frac{1}{2}\ln(1-2x)$ gives

$$\varphi_2(t) = \psi_2^{-1}(t) = \frac{1 - e^{-2t}}{2},$$

an affine (hence Möbius) image of $\exp(-2t)$ — family $\exp(\alpha t^\rho)$ with $\rho = 1$, $\alpha = -2$. Correspondingly the substitution $u = 1 - 2x$ conjugates f_2 to the monomial $u \mapsto u^2$, and the linearised fixed point has multiplier 2 (repelling), as theorem 3.11(i) requires.

Case $r = 4$. Inverting (27): $t = (\arcsin\sqrt{x})^2$ gives

$$\varphi_4(t) = \psi_4^{-1}(t) = \sin^2\sqrt{t} = \frac{1 - \cos(2\sqrt{t})}{2},$$

a Möbius image of $\cos(\alpha t^{\rho}+\beta)$ with $\rho=\frac{1}{2}$, $\alpha=2$, $\beta=0$ — the fractional exponent is admissible precisely because $\cos(2\sqrt{t})$ is an entire function of t. Correspondingly $u=1-2x$ conjugates f_4 to $T_2(u)=2u^2-1$, the second Chebyshev polynomial, and the multiplier at the linearised fixed point is 4 (repelling).

Both examples therefore sit exactly where theorem 3.11 says all differentially algebraic Schröder functions must sit: over repelling fixed points, in the exponential and cosine families. For $r\in(0,1)$ the fixed point is attracting, no member of the classification is available, and theorem 3.13 gives hypertranscendence of ψ_r — which underpins the main-text claims for $A(p)=2\psi_{2p}(\frac{1}{2})$.

.2 Closed-form searches and quadratic conjugacy

This appendix records the negative and structural calculations that are useful for interpreting $A(p)$, but are not needed for its probabilistic role in the main text. The numerical formulae below are approximations only. They are retained because they show how little a visually accurate fit says about the singular quantity $1/A(p)$.

.2.1 Exploratory formulae

Polynomial and rational searches can be constrained to the known endpoints $A(0)=1/2$ and $A(1/2)=0$. One of the more economical rational candidates was

$$\widehat{A}_1(p)=\frac{\frac{1}{2}(1-2p)(1+2p)}{1+\frac{1}{2}p-\frac{5}{2}p^2-\frac{6}{5}p^3+\frac{6}{5}p^4}. \tag{28}$$

It passes through both endpoints. At the critical endpoint, however,

$$\widehat{A}_1'(1/2)=-\frac{2}{0.55}\approx -3.636, \tag{29}$$

whereas theorem 3.4 requires the left derivative -4. Matching the values but missing this slope is enough to make the relative error in A, and hence the error in $1/A$, unacceptable near criticality.

A less restricted symbolic tree search produced many longer formulae of similar visual quality. Two representative outputs were

$$\begin{aligned}\widehat{A}_2(p)=0.31696448624690005\log\Bigg(\Bigg|\cos\Bigg(\Big(\frac{1}{3.19801133824532}\Big)^{1/4}\\ +p\Big[\sin\Big(\cos\big(|p|(0.621865696665606-p)\cos(\mathrm{e}^{\mathrm{e}^{p}})\big)\Big)+|p|\Big]\Bigg)\Bigg|\Bigg)+0.5982282661889831\end{aligned} \tag{30}$$

and

$$\widehat{A}_3(p)=\frac{-0.616647881739505\,\mathrm{e}^{-0.552605335919693\,\mathrm{e}^{p}}}{\sin(\cos p)-|p|}+0.921964323679771. \tag{31}$$

These expressions have no inferential status: they are records of a curve-fitting exercise, not proposed identities.

One can force the correct critical branch by splicing:

$$\widehat{A}_4(p)=\begin{cases}-0.04011483983619545\,\mathrm{e}^{\mathrm{e}^{2p}}+0.6079994336528085, & p<0.499962,\\ 2-4p, & p\geq 0.499962.\end{cases} \tag{32}$$

This is a practical surrogate, not a closed form for A. If a splice is needed, the certified product in (3.9) is preferable on the first branch and the proved asymptotic (3.15) on the second.

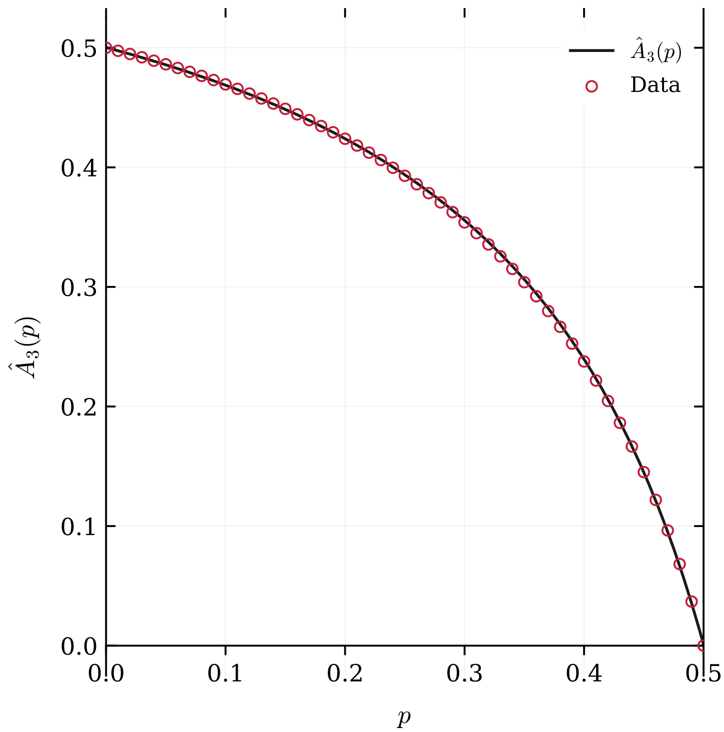

Figure 9: The fitted curve $\widehat{A}_3$ and product evaluations of $A(p)$. The discrepancy is barely visible on this scale, although $\widehat{A}_3(1/2) \approx 9.1 \times 10^{-4}$ rather than zero and its endpoint slope is approximately -3.63 rather than -4.

.2.2 Conjugacy with the quadratic family

The logistic family $f_r(x) = rx(1-x)$ is affinely conjugate to the quadratic family

$$P_c(z) = z^2 + c. \tag{33}$$

Set

$$z = r\left(\frac{1}{2} - x\right). \tag{34}$$

If $x' = f_r(x)$ and $z' = r(1/2 - x')$, direct substitution gives

$$z' = z^2 + c, \qquad c = c(r) := \frac{2r - r^2}{4}. \tag{35}$$

The subcritical branching window $0 < r < 1$ therefore maps onto $0 < c < 1/4$.

The interval $0 < c < 1/4$ is the real radius of the main cardioid of the Mandelbrot set. Indeed, a fixed point of $z^2 + c$ with multiplier r has coordinate $z_* = r/2$, and its fixed-point equation gives

$$c = z_* - z_*^2 = \frac{r}{2} - \frac{r^2}{4}. \tag{36}$$

Thus the endpoint $r \uparrow 1$ approaches the parabolic cusp $c = 1/4$. This observation locates the parameter range; it is not used as evidence for a closed-form claim.

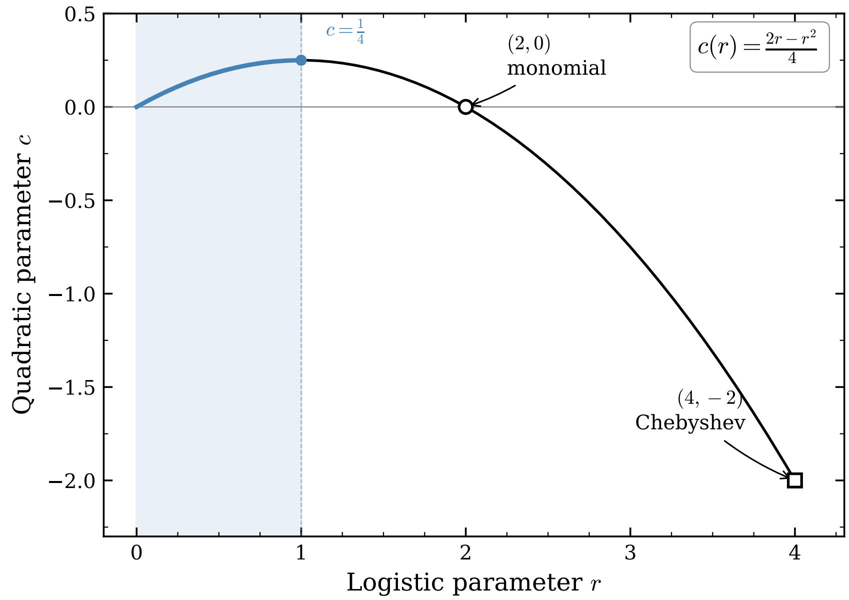

Figure 10: The parameter correspondence $c(r) = (2r - r^2)/4$. The subcritical window $0 < r < 1$ maps to $0 < c < 1/4$; the elementary monomial and Chebyshev parameters described below lie outside that window.

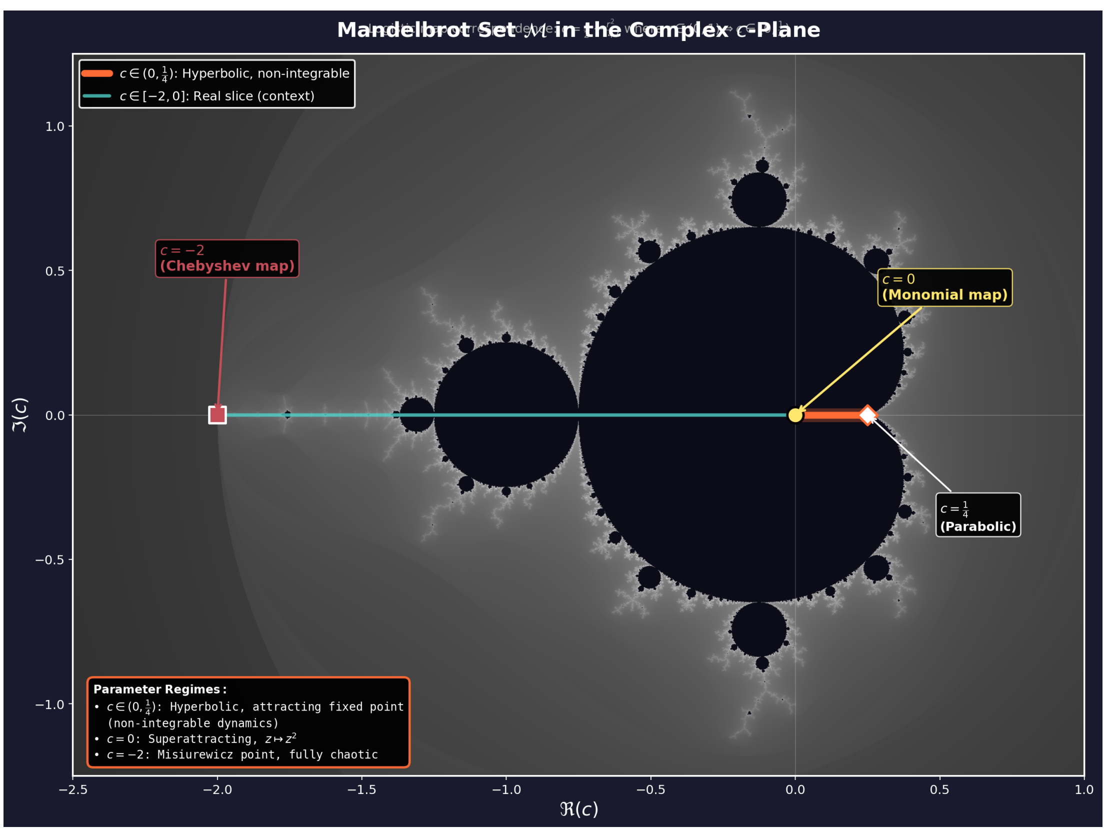

Figure 11: The image $0 < c < 1/4$ of the subcritical branching interval within the quadratic parameter plane. The exceptional parameters $c = 0$ and $c = -2$ are shown for comparison.

.2.3 The two elementary comparison cases

The exceptional quadratic maps conjugate to a monomial or Chebyshev polynomial correspond to elementary linearisers. They furnish useful algebraic checks, but neither belongs to $0 < r < 1$.

For $r = 2$, put $u = 1 - 2x$. Then

$$1 - 2f_2(x) = (1 - 2x)^2 = u^2, \tag{37}$$

and the normalised local lineariser is

$$\psi_2(x) = -\frac{1}{2}\log(1 - 2x). \tag{38}$$

It satisfies $\psi_2(f_2(x)) = 2\psi_2(x)$, $\psi_2(0) = 0$ and $\psi_2'(0) = 1$.

For $r = 4$, write $x = \sin^2\theta$. The double-angle identity gives

$$f_4(x) = 4x(1 - x) = \sin^2(2\theta), \tag{39}$$

so that

$$\psi_4(x) = \{\arcsin(\sqrt{x})\}^2 \tag{40}$$

satisfies $\psi_4(f_4(x)) = 4\psi_4(x)$ and $\psi_4(x) \sim x$ at the origin.

There is a categorical distinction here. At $r = 2$ and $r = 4$, the origin has multiplier greater than one and is repelling; the formulae (26) and (27) are local repelling linearisers. In the branching window, the origin is attracting and the function in (3.20) is the attracting Koenigs coordinate. The elementary formulae are therefore comparisons, not analytic continuations of $A(p)$.

Finally, the theorem cited in section 3.8 concerns the function $z \mapsto \psi_r(z)$ at fixed r. It does not prove that the parameter evaluation $p \mapsto 2\psi_{2p}(1/2)$ is non-elementary. This logical separation is the material correction to the stronger claim suggested by the original closed-form search.

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